Earth Moving Algorithm Performance Analysis

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Executive Summary

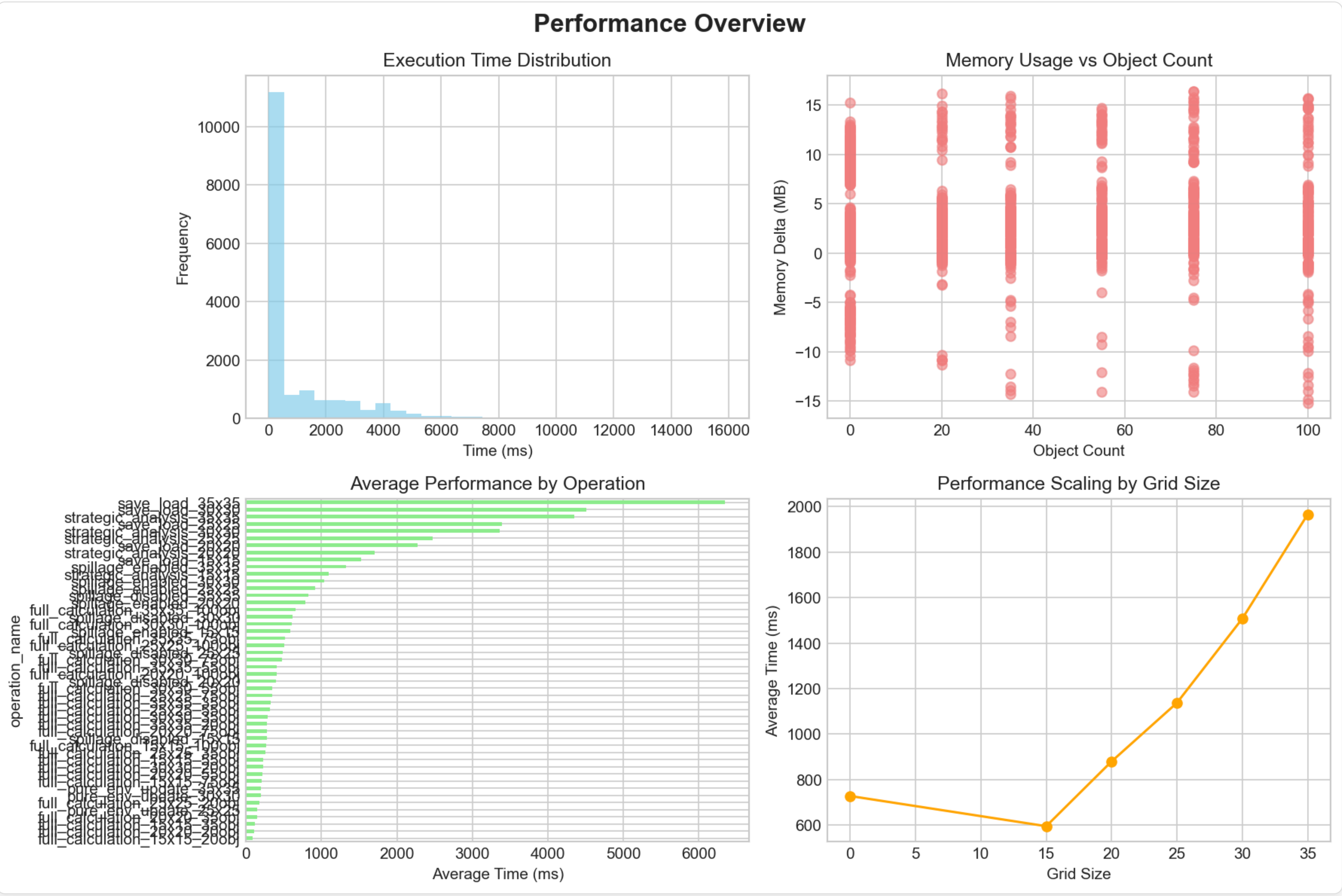
Overall Performance	Test Coverage	Scalability
Needs Optimization	**16,470 samples**	**Variable**
Average: 1007.9ms ± 1529.6ms	2161 scenarios tested	Complex/Decreasing - Unusual behavior

Key Findings

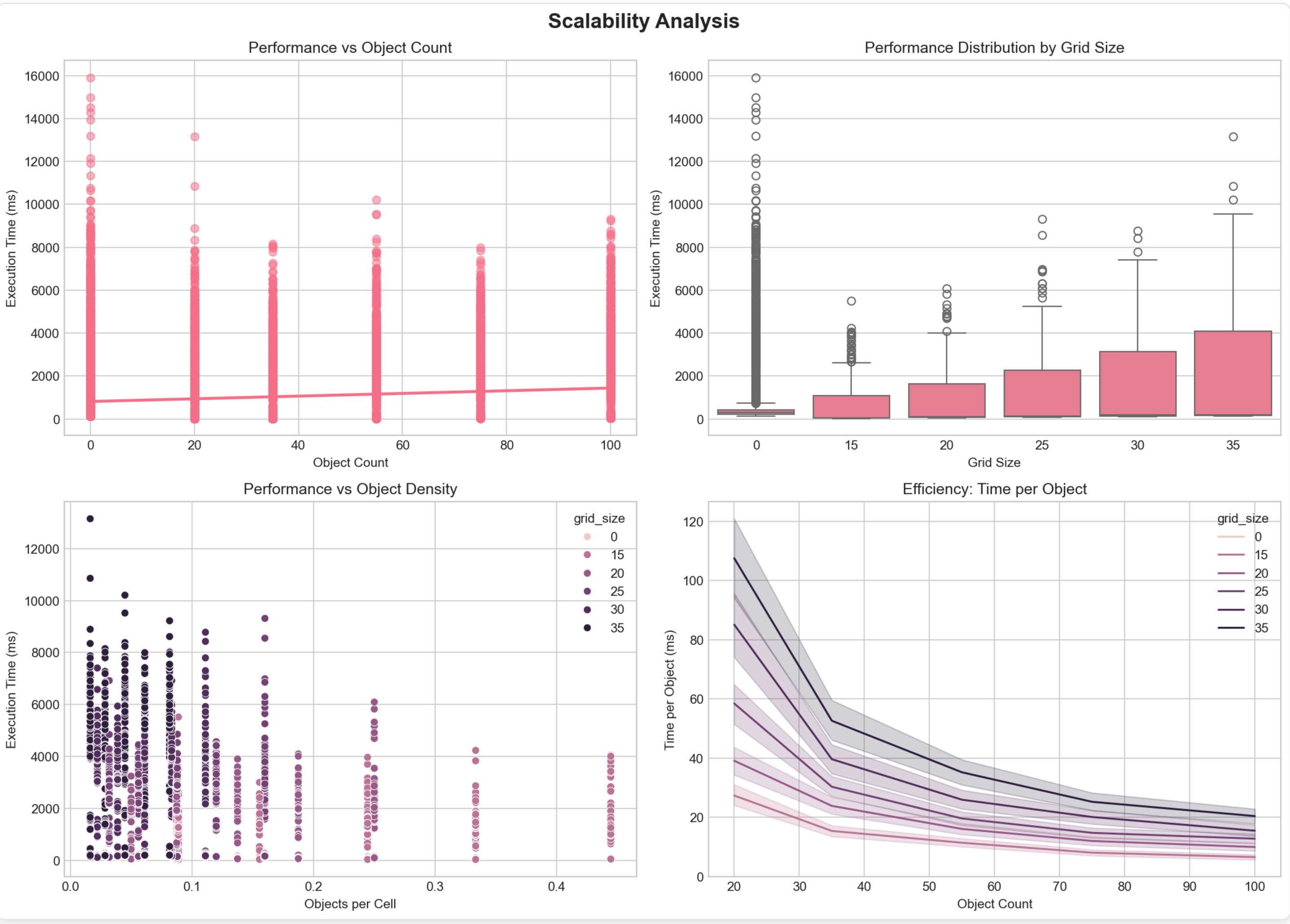
→ Spillage mode adds -63.4% performance overhead
→ Algorithm exhibits complex/decreasing - unusual behavior scaling behavior
→ Performance optimization recommended for large scenarios

Performance Visualizations

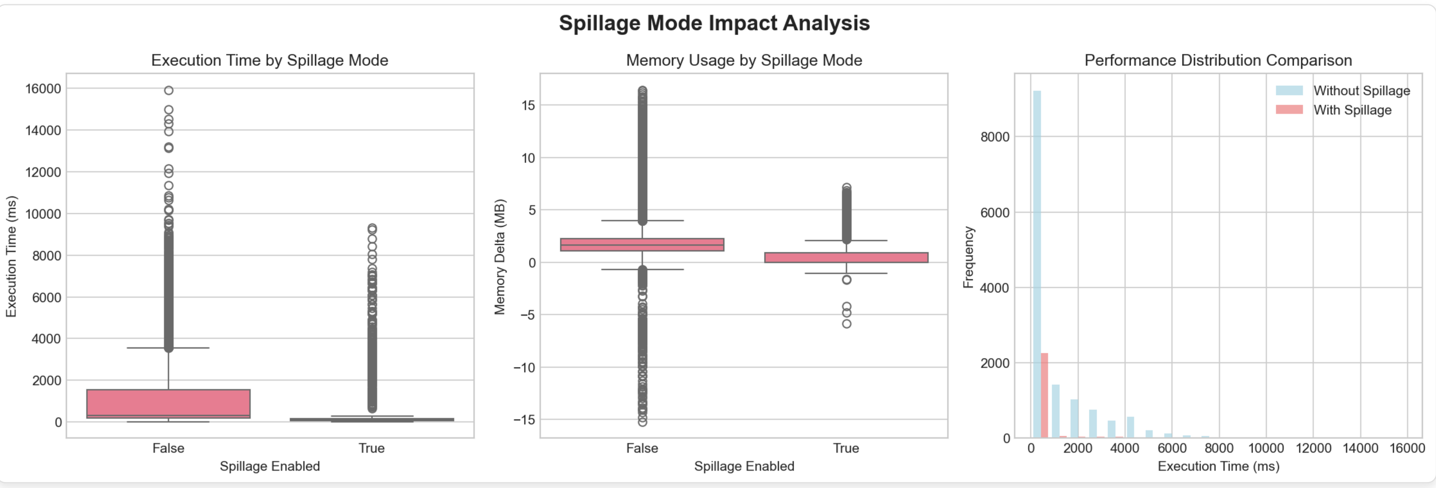
Performance Overview



Scalability Analysis



Spillage Impact Analysis



Detailed Performance Analysis

Scalability Analysis

Grid Size

Complexity: O(n²) - Quadratic with linear component

Linear Fit R²: 0.702

Predictability: Moderate

Object Count

Complexity: Complex/Decreasing - Unusual behavior

Linear Fit R²: 0.528

Predictability: Low

Total Cells

Complexity: O(n²) - Quadratic with linear component

Linear Fit R²: 0.931

Predictability: High

Spillage Mode Impact

Performance Overhead: -63.4%

Recommendation: Low overhead - spillage recommended for better accuracy

Statistical Significance: No

Operation Performance Comparison

Operation	Average Time (ms)	Std Dev (ms)	Min (ms)	Max (ms)	Samples
Full Calculation 15X15 20Obj	85.38	208.67	16.01	1150.15	200
Full Calculation 15X15 35Obj	119.76	305.24	21.19	1770.62	200
Full Calculation 15X15 55Obj	226.31	656.24	26.87	3966.15	200
Full Calculation 15X15 75Obj	211.52	650.30	28.89	4231.97	200
Full Calculation 15X15 100Obj	264.53	805.47	34.19	4059.01	200
Full Calculation 20X20 20Obj	110.22	189.98	42.41	1004.86	200
Full Calculation 20X20 35Obj	147.31	308.26	47.35	1572.62	200
Full Calculation 20X20 55Obj	214.03	508.06	52.23	2859.60	200
Full Calculation 20X20 75Obj	281.75	760.41	57.61	4094.47	200
Full Calculation 20X20 100Obj	405.46	1134.41	61.78	6092.68	200
Full Calculation 25X25 20Obj	183.12	277.54	75.17	1627.64	200
Full Calculation 25X25 35Obj	254.61	472.54	79.72	2412.19	200
Full Calculation 25X25 55Obj	322.29	716.25	87.06	3697.44	200
Full Calculation 25X25 75Obj	347.85	856.16	87.91	4193.01	200
Full Calculation 25X25 100Obj	511.32	1439.70	93.09	9316.44	200
Full Calculation 30X30 20Obj	224.06	314.01	106.05	1572.37	200
Full Calculation 30X30 35Obj	283.57	469.08	111.69	2259.64	200
Full Calculation 30X30 55Obj	350.01	711.56	116.37	3675.11	200
Full Calculation 30X30 75Obj	474.14	1074.89	122.33	5312.36	200
Full Calculation 30X30 100Obj	607.02	1605.32	126.48	8782.09	200
Full Calculation 35X35 20Obj	282.52	384.67	135.65	1733.55	200
Full Calculation 35X35 35Obj	328.74	509.52	142.43	2752.51	200
Full Calculation 35X35 55Obj	409.00	765.68	153.97	3555.95	200
Full Calculation 35X35 75Obj	514.17	1046.05	159.07	5659.81	200
Full Calculation 35X35 100Obj	658.10	1596.46	167.89	9224.96	200
Pure Env Update 25X25	150.79	14.94	131.68	256.02	240
Pure Env Update 30X30	193.49	46.22	158.71	616.26	240
Pure Env Update 35X35	201.53	34.68	179.52	535.22	240
Spillage Enabled 15X15	584.28	980.13	133.04	5083.50	750
Spillage Disabled 15X15	281.50	249.49	133.37	1665.42	750
Spillage Enabled 20X20	786.20	1381.55	181.87	8005.24	750
Spillage Disabled 20X20	401.22	421.03	180.95	5317.75	750
Spillage Enabled 25X25	917.48	1577.22	228.99	9388.64	750
Spillage Disabled 25X25	486.63	454.01	228.88	2463.73	750
Spillage Enabled 30X30	1038.06	1718.83	273.44	9686.65	750
Spillage Disabled 30X30	622.01	663.03	274.80	4595.60	750
Spillage Enabled 35X35	1323.36	2331.93	331.36	15914.41	750
Spillage Disabled 35X35	831.49	983.27	332.51	7242.86	750
Strategic Analysis 15X15	1097.09	186.82	949.22	2831.39	450
Strategic Analysis 20X20	1701.09	277.49	1469.19	3701.32	450
Strategic Analysis 25X25	2469.16	369.81	2127.37	4433.12	450
Strategic Analysis 30X30	3364.01	514.72	2957.82	7406.55	450
Strategic Analysis 35X35	4353.08	506.22	3884.51	7293.78	450
Save Load 15X15	1529.02	489.95	1197.25	5517.52	200
Save Load 20X20	2269.52	440.92	1850.13	3901.13	200
Save Load 25X25	3389.83	681.29	2747.53	8553.41	200
Save Load 30X30	4507.59	586.75	3844.61	7300.85	200
Save Load 35X35	6350.15	1162.09	4984.78	13154.19	200

Recommendations

- Consider performance optimization for scenarios exceeding 1 second execution time
- Spillage Mode: Low overhead - spillage recommended for better accuracy